

Exploring Emotional Reappraisal and Repression through Acoustic Mood Self-Tracking

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ABSTRACT

Self-tracking technologies are a dominant tool employed in Human-computer Interaction (HCI) in the realm of mental health research. The majority of these tools serve as mechanisms for self-reflection in individuals with affective disorders and are often used to inform medical treatment provided by caregivers. Mood tracking, in particular, has the potential to be expanded into other explorations beyond mere trend-lines of affective states; behaviors such as emotion reappraisal and repression could be measured and influenced as a means of improving user mental health outcomes. As self-tracking technologies face numerous challenges in the motivation and engagement of persons with mental illnesses, this position paper forwards an alternative approach demonstrated through “TENOR,” an acoustic mood-tracking application designed to increase user involvement through music creation and emotional recall.

CCS CONCEPTS

• Human-centered computing → Usability testing.

KEYWORDS

Self-Tracking Technology, Mental Health, Emotions, Emotional Appraisal, Emotional Suppression, Emotion Regulation

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1 INTRODUCTION

Self-tracking technology can trace some of its roots back to the Quantified Self (QS) movement that emerged in 2007, when notions of an ideal and “optimized” version of ourselves were said to be achievable through a *datafication* of our entire personhood [14]. Many have been quick to criticize the idea and refer to it as **data**

fetishization, seeing as it encourages the belief that data is superior and objective in mapping out trajectories of self-improvement, which runs contrary to the nonlinear and haphazard nature of the act itself [25].

The idealization of a maximal “self” also suggests that there is an inherent deficiency in people that must be remedied to fit externally-imposed standards of *good* and *healthy*, assuming normative ideals for what it means to maximize these states [25]. This issue has its aftershocks laid out in the consequences of self-tracking technologies existing today, as many of them have shown to worsen the condition of users once their performance evaluations fail to match the hard work and sincerity put into their efforts [12, 23]. Binary conceptions of “good” and “bad” could risk alienating the diversity of the human experience, which enjoys a range of grays just as worthy of exploration without a pressure to conform [8]. In the case of mood-tracking, a large portion of prevailing applications accessible to the public make the same mistake: Platforms automatically assign positive emotions as “good” while negative emotions are “bad,” and reinforce these perceptions through the use of pictures, words, and colors as classification systems [5]. Conversely, positive emotions can indicate undesirable symptoms in some affective disorders [24], calling into question the integrity and universality of these standards. In fact, even mixed and negative emotions have shown to benefit individuals when processed and prescribed meaning to, as the experience offers them the strength and resolve to confront difficult situations in the future [11].

There is a need to introduce mood-tracking systems that do away with measuring performance through social conventions. Self-improvement is a social endeavor, as individuals develop themselves through the influence of others [20], and the manner in which one manages their emotional regulation has its own social cost and benefits [9]; however, these factors should foster positive changes as opposed to intimidating one to comply in order to fit a certain mold and reach acceptance. For this reason, this paper proposes an alternative emotion self-tracking technique focused on creative expression. “TENOR” capitalizes on music creation to categorize emotions while focusing on the user’s individual actions of emotional repression and reappraisal. This ensures that any self-improvement is defined by the user, who alone can determine whether they are approaching their desired emotions with greater acuity.

2 BACKGROUND AND FOUNDATIONS

Digital mood-tracking techniques range from wearable technology and behavioral monitoring to self-reporting systems [16]. While the first two offer compelling non-invasive longitudinal monitoring, the most accessible of the three is electronic self-reporting; several

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mood-tracking applications designed for a general audience can be downloaded for free from smartphone app stores. Neurotypical individuals primarily make use of these platforms to determine trends and patterns in their moods. However, those with affective disorders are just as likely to choose and prefer general-use mood-tracking applications over ones specifically catered to their illness; this is due to their greater flexibility in tracking a variety of symptoms [5].

This underscores the importance of identifying the unique needs and preferences of those with affective disorders and how the manifestations of their diagnoses affect their lifestyle and welfare: Symptoms of a mental illness can lead to an aversion to technology use [18] and reading long material [7], as well as an inability to maintain a regular tracking routine [19]. Moreover, defining improvement in emotional disposition through normative perceptions of "good" and "bad" disregards the individualized complexity of mood fluctuations, most especially in regards to affective disorders; excessive positivity, for example, could be a sign of an oncoming manic episode for those with Bipolar Disorder [24].

2.1 The Anatomy of Mobile Mood-Tracking Applications

Mobile mood-tracking applications for general use typically cover four stages: **Preparation**, **Collection**, **Reflection**, and **Action** — Preparation refers to instructions on how one should track their mood, Collection pertains to the act of inputting mood data, Reflection features visualizations such as graphs and charts outlining mood trends, and Action prescribes recommendations on how to improve one's mood based on the data presented [5]. In a feature analysis surveying the characteristics of 32 mobile mood-tracking applications, majority were found to fall short on **Preparation** and **Action** features: Very few applications presented the user with an incentive for mood-tracking while only 7 out of 32 applications offered methods or suggestions to help improve mood [5].

It is evident that users require more "priming" regarding the benefits of mood-tracking in order to build and maintain the motivation to sustain the habit. This, along with the tendency of individuals with affective disorders to struggle with routines [19], underscore a demand to make mood-tracking more engaging.

2.2 Confronting Unaddressed Emotional Phenomena

Absent from the conversation on mood-tracking are several factors intrinsic to the emotional experience: The **Fading Affect Bias (FAB)**, which pertains to the faster decaying of intensity in negative emotions as compared to positive emotions [22], and the practice of emotional **Reappraisal** and **Suppression** in emotion regulation [9]. Both FAB and emotion suppression serve as self-preservation tactics that protect individuals from reliving traumatic events [3, 10, 21, 27], however, habitual use of emotion suppression can pose detrimental effects to relationship satisfaction and come with a social cost [9]. Helping users replace suppression with acceptance can lead to more efficient coping strategies [1]. These three facets stand to be addressed and explored by mood-tracking applications as a means of increasing user involvement in the process of

confronting their emotions; doing so may even provide users with the direction needed to improve their well-being.

3 OVERVIEW OF TENOR

In a bid to explore interventions for the gaps found in mood-tracking applications, this paper proposes an alternative tracking technique that employs a creativity-based approach to emotion classification. TENOR is mood-tracking tool that encourages users to translate their overall day-to-day emotional experiences through interactions with a 256-key virtual drum pad. The tool endeavors to complete the four stages of mood-tracking through its user flow: Users are prompted to create songs on the interface in whatever way they wish, giving them free reign to map out their affective states in ways only they are most suited to interpret [**Preparation** and **Collection**] (Fig. 1). This does away with the latent issue of pushing normative ideas of "positive" and "negative" emotions upon the user, as the meaning of these emotional valences are instead left for them to define for themselves. Afterwards, users are encouraged to select items from an emotional word bank to attach to their creations (Fig. 2), which are then used at the end of the week to gauge their memory retention of past emotions when prompted by listening to the song they created [**Action**] (Fig. 4). This feature works to persuade the user to reappraise previous sentiments and increase their emotional acuity through deeper introspection. Having a means of quantifying user inputs for the sake of testing requires that they once again be subjected to another model and additional tasks which may be disruptive. Future studies will seek ways to diminish this as much as possible. In addition to these features, users can revisit their creations during any time in the process through a song listing that outlines their weekly inputs [**Reflection**] (Fig. 3).

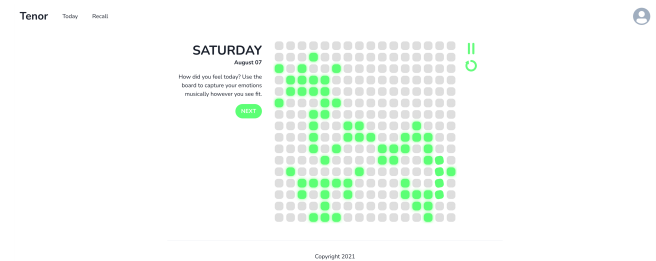


Figure 1: TENOR Music Grid

3.1 Design Paradigm

In order to give context to the final form of TENOR, here we briefly describe the platform's design paradigm. Balance and symmetry in interfaces induce impressions of usability and minimize the amount of attentional resources demanded from users, which is essential to maximizing focus and concentration [4, 26]. At the same time, light or dark gray colors are ideal for directing attention towards the content of an interface [2]. These concepts largely informed the design and aesthetics of TENOR (Fig. 1), as priority was placed upon

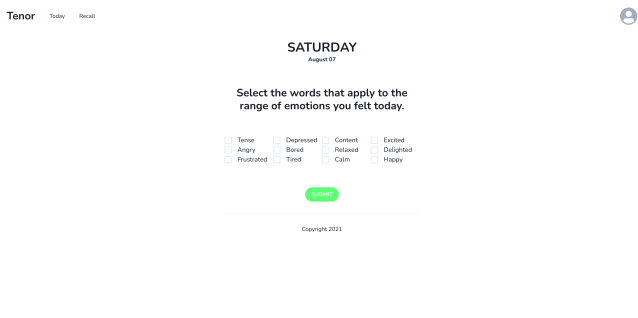


Figure 2: TENOR Emotional Word Bank

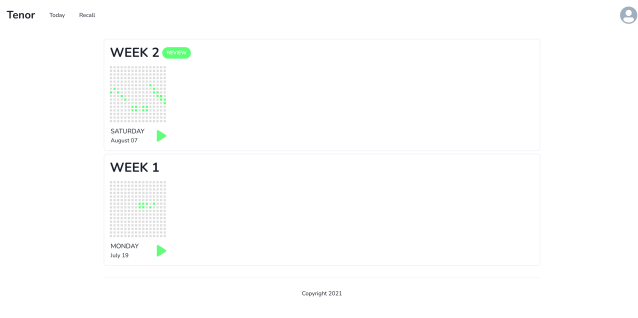


Figure 3: TENOR Song list

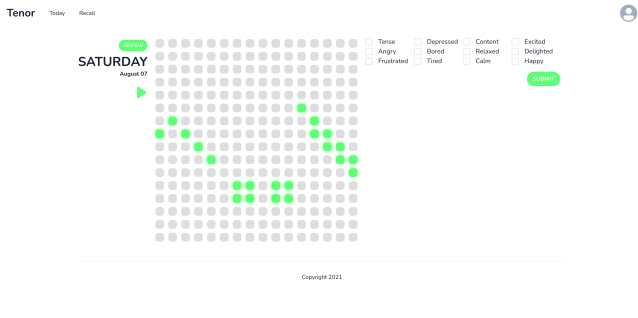


Figure 4: TENOR Recall Test

ensuring that the appearance, features, and action points presented to the user do not overwhelm them and subsequently intimidate them.

Certain colors invite notions of emotion: Red, orange, and yellow are often attributed with arousal or exuberance while shades of blue indicate placidity and quietude [15, 17]. Comparatively, green has shown to aid in memory retention of positive information [13] and invokes a sense of calm and trustworthiness [6]. To diminish the amount of emotional associations that could be made from the interface design, green was used for its general neutrality and its ability to positively impact memory.

3.2 Giving Dimension to Emotion Classification

The emotional word bank featured on TENOR’s platform makes use of a 12-item two-dimensional valence and arousal matrix. The four quadrants represent **High-Positive (I)**, **High-Negative (II)**, **Low-Negative (III)**, and **Low Positive (IV)** affective states (Fig. 5). Given the changeable nature of emotional arousal [22], determining the extent in which its fading intensity affects reappraisal and recall can aid in determining the particular emotions users have trouble confronting.

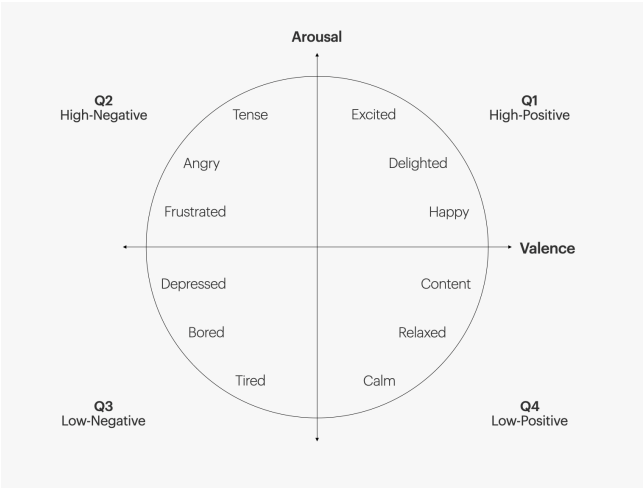


Figure 5: Two-Dimensional Valence and Arousal Matrix. Based on matrix from Predicting Valence-Arousal Ratings of Words Using a Weighted Graph Method by Yu et al. [28]

4 TESTING AND OBSERVATIONS

To determine whether TENOR’s music interface increases engagement and recall when tracking moods, a pilot test was conducted with 17 participants over a two week period. Users interacted with two different test modes each week and in a randomized sequence: **Control** (tracking with an emotional word bank only) and **Experimental** (tracking with the music interface and word bank). 9 users began with the treated setup on the first week and the untreated setup in the second week while 8 users experienced the opposite. From Monday to Wednesday, users log their emotions into the platform and then spend the three following days away on a period of rest. They return on Sunday to respond to a recall test prompting them to remember their inputs from the most recent tracking day: Wednesday. The importance of the recall portion of the application lies in our objective to encourage emotional reappraisal in users; when individuals are prompted to remember past feelings, they may be driven to reexamine them on a deeper level. Future experiments may seek out ways to measure the levels of introspection done by users voluntarily, or may incorporate additional prompts that lead users to evaluate their emotions after recalling them.

In order to investigate the impact of test mode on the recall quality of emotions with respect to their valence quadrant, we constructed a mixed effects model where recall instances were broken

Table 1: Pairwise Test Results

1	Estimate	p.value
0-1	-3.42202	0.0037
0-2	-1.56548	0.0432
0-6	-1.68209	0.0296
1-3	2.25845	0.0433
1-4	2.21139	0.0492
1-7	2.29272	0.0414

Table 2: Quadrant and Mode Combinations

No.	Combination	No.	Combination
0	Quadrant 1 + Control	4	Quadrant 1 + Experimental
1	Quadrant 2 + Control	5	Quadrant 2 + Experimental
2	Quadrant 3 + Control	6	Quadrant 3 + Experimental
3	Quadrant 4 + Control	7	Quadrant 4 + Experimental

up into eight groups: The four valence and arousal quadrants for each testing mode (Table 2). We then fit a mixed-effects model with correctness as binary dependent variable and the eight quadrant/testing scenarios as independent variables. Participants were a random factor.

The model showed that three of the quadrants were significant against the dummy variable. The pseudo R-squared is 0.13, which is small. However, it shows that valence and mode can have an impact on correctness.

To get a general idea of impact of quadrants and testing mode, we ran a pairwise test without adjustment (Table 1). The decision to not use a Tukey post-hoc adjustment was intentional due to the number of comparisons and low sampling power. These results should be seen as preliminary and are used to justify future research.

4.1 Insights

The results of the pairwise test revealed six significant pairs: Q1 in Control v. Q2 in Control [0,1], Q1 in Control v. Q3 in Control [0,2], Q2 in Control v. Q4 in Control [1,3], Q1 in Control v. Q3 in Experimental [0,6], Q2 in Control v. Q1 in Experimental [1,4], and Q2 in Control v. Q4 in Experimental [1,7] (Fig. 6).

Users performed better when recalling high-negative emotions in the Control setup as opposed to both high and low positives in Experimental. This is likely due to a dampening of negative arousal states caused by the delight attached to using and revisiting the music interface, which is absent in the word bank mode. On the other hand, low-negative emotions were recalled significantly better in the Experimental setup in comparison to high-positives in Control; this demonstrates the capacity of the music interface to engender low-arousal negative emotions in users despite its enjoyment factor.

The absence of significant comparisons between Quadrant-Mode (QM) combinations within the music interface reveals its stability in regards to encouraging recall in users—participants were consistent in their reappraisal of past emotions due to the prompting that their music creations provided them. In contrast, QM combinations

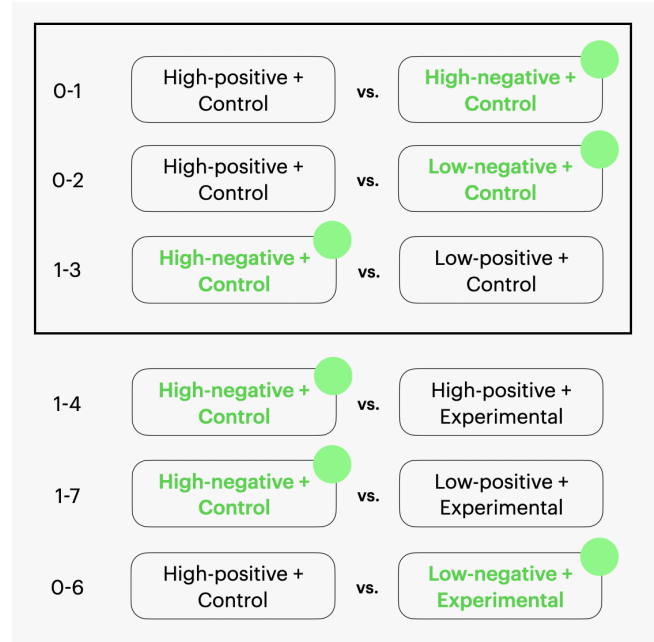


Figure 6: Pairwise Test Results. The green dots refer to the better performing quadrant and mode combination within each pair. The box highlights the discrimination within the control setup.

suffered and competed with one another within the word bank group, propounding high discrimination and fluctuations in recall quality in the Control setup.

While it is more difficult to remember negative valences in the Experimental mode, the music interface still shows greater promise overall in inducing recall across the board—users are more attuned to emotional reappraisal when using an engaging interface that prioritizes their involvement. The next step in this experiment would be to determine ways to reduce the emotional dampening caused by the enjoyment of interacting with the interface. That said, there is potential to uncover more insights through further testing with a larger participant pool given the outcomes in the pairwise test.

5 CONCLUSION

This paper posits a more interventional approach to mood-tracking that encourages emotional reappraisal in users while minimizing repression. Rather than make use of words, colors, and pictures to categorize emotions, TENOR has users interact with a virtual drum pad and generate songs emblematic of their affective states. This increases the user's involvement in the tracking process, giving them incentive to build a routine and confront their emotions with less intimidation. The use of individual music creations to classify moods also gives users the license to define and interpret their emotions on their terms without feeling pressured to conform to externally-imposed standards.

A pilot test conducted to determine the effectiveness of TENOR's interface as opposed to an emotional word bank revealed greater

recall efficiency and consistency among all mood valences across the board. This demonstrates the benefits of increasing user involvement in the mood-tracking process; users are more in touch with their emotions and gain the opportunity to process them through reappraisal and introspection. Future studies should go into mitigating the arousal decay of negative emotions caused by the interface's delight factor, and further insights could be gained from conducting tests with a larger participant pool that accounts for potential sociocultural differences in emotion regulation.

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